E-Commerce Sales & Customer Analytics

Corrected Business Insights & SQL Review

This version is based on the SQL script provided by the project owner and the results previously reported in the conversation. It separates SQL-supported analysis from reported findings and flags calculations that must be validated before public release.

1. Project Overview

This project uses MySQL to analyze orders, order items, customers, products, payments, and returns, and Power BI to present KPIs and interactive visuals. The SQL script includes analysis of revenue and AOV, monthly sales, fulfilment status, product/category performance, geographic sales, customer segments and repeat purchasing, delivery times, returns, payments, inventory, and discount levels.

2. Regional Finding — Andhra Pradesh

Reported finding from the project owner: Andhra Pradesh was identified as the highest-performing state for orders and revenue in the analysis. This is an important project-specific insight and should be featured in the executive summary and Power BI discussion.

The supplied SQL includes state-level revenue/order queries and separate category analyses for Andhra Pradesh and Telangana. However, the pasted script contains the query text, not the output rows. Therefore, the exact AP order count, AP revenue, Telangana comparison values, and ranking across all states cannot be independently recalculated from the script alone. Do not invent exact AP values in the PDF; insert the actual result-grid values if available.

Important SQL grain correction for state/category queries

When joining orders to order_items, one order may appear on multiple item rows. In the AP/Telangana query, COUNT(o.order_id) counts joined item rows, not necessarily unique orders. For unique orders use COUNT(DISTINCT o.order_id). SUM(oi.line_total) is suitable for item/category revenue, provided line_total reflects the project's intended revenue definition.

3. Key Metrics Previously Reported

Metric	Previously reported value	Review status
Total orders	60,000	Reported earlier; verify with COUNT(*) from orders.
Total order revenue	24,087,159.91 (≈24.09M)	Reported earlier; based on SUM(orders.total_payable). Verify total_payable has no NULLs and definition is consistent.
Average order value	≈401.45	Consistent arithmetically with 24,087,159.91 / 60,000, but rerun the query to confirm.
Delivered orders	41,960 (≈69.93%)	Reported earlier; order-count rate.
Cancelled orders	3,689 (≈6.15%)	Reported earlier; order-count rate.
Returned orders	2,374 (≈3.96%)	Reported earlier; order-count rate.
Delivered revenue	≈16,863,163.64; ≈70.01% of total	Reported earlier; revenue share, not order-count share.
Payment success rate	82.48%	Reported earlier; validate denominator and duplicate payment records.
Hyderabad Electronics revenue	618,000	Reported earlier by project owner; confirm the underlying query and whether this is line revenue or order-level revenue.

These are previously reported results, not outputs embedded in the SQL file. Before publishing, rerun the queries and confirm all figures, definitions, and currency.

4. Useful Business Insights Supported by the Project Work

A. Regional performance

Andhra Pradesh was reported as the top state for orders and revenue. The SQL also analyzes category revenue within Andhra Pradesh and Telangana. Use this finding to discuss regional demand, then use category-level outputs to identify which categories contribute most within each state. Exact AP totals and the AP-versus-Telangana comparison must be taken from the actual query results.

B. Order fulfilment

Previously reported order counts show 41,960 Delivered, 3,689 Cancelled, and 2,374 Returned out of 60,000 total orders. This corresponds to approximately 69.93%, 6.15%, and 3.96%, respectively. These are status shares by order count. Delivered revenue share (reported at 70.01%) is a separate metric.

C. Discount and AOV

Previously reported AOV values by discount were: 0% = 443.12; 5% = 426.85; 10% = 405.09; 15% = 385.31; 20% = 365.43. In these results, higher discount groups had lower average order values. This is an observed association and does not establish that discounts cause lower AOV; product mix and customer/order characteristics may differ.

D. Inventory watchlist

Previously reported examples include Yoga Mat Classic (stock 0, historical delivered units sold 76) and Headphones Pro (stock 5, historical delivered units sold 81). These are potential replenishment signals, not a forecast. Check recent sales, seasonality, supplier lead time, and incoming stock before making purchasing decisions.

E. Payments and returns

The script includes payment status counts, payment success, payment method analysis, return reasons, and refund status. The reported payment success rate was 82.48%, but the denominator must be stated (payment records versus unique orders). The return-reason queries can identify common reasons, but the actual counts are not included in the pasted script.

F. Customer retention

The script includes repeat-purchase analysis and customers who registered but never placed an order. A repeat-purchase rate is only interpretable when its denominator is explicit: repeat customers divided by all registered customers is different from repeat customers divided by customers who placed at least one order. Recalculate and label the intended definition before publishing the rate.

5. SQL Review — Corrections Before Publishing

1. Revenue duplicated after joining order_items: Queries that join orders to order_items and then SUM(o.total_payable) may multiply order revenue when an order has multiple item rows. Use orders alone for order-level revenue; use SUM(oi.line_total) for item/category revenue.

2. Unique orders in AP/Telangana category queries: Change COUNT(o.order_id) to COUNT(DISTINCT o.order_id) if the intended metric is unique orders. Otherwise label it as item rows/order-item occurrences, not orders.

3. Monthly growth: The first month uses LAG(..., 1, 0), then divides by the previous value. A previous value of zero should produce NULL/not applicable, not a meaningful growth percentage. Use LAG(total_sales) and NULLIF(previous_month_sales,0).

4. Delivered order count growth: Joining orders to order_items and COUNT(*) counts item rows, not orders. Use COUNT(DISTINCT o.order_id) for order count.

5. Top 10 delivered products: The query groups by product_name but includes order_status in SELECT without grouping/aggregation. Remove order_status (it is already filtered to Delivered), group by product ID and product name, and decide whether ranking is by revenue, units, or order count. The dashboard title says Top 10 by Revenue, so sort by total_revenue DESC if that is the intended ranking.

6. Payment method/status join: If an order has multiple payment records, joining payments to orders can duplicate order revenue. Use payment-level measures for payment records and deduplicated order-level aggregates for order revenue.

7. AOV and NULL total_payable: COUNT(order_id) counts orders even if total_payable is NULL, while SUM ignores NULLs. Check for NULL total_payable and define the denominator accordingly.

8. Delivery time: The average query is item-level because it uses order_items. If the business metric is average delivery time per order, aggregate delivery dates to one order-level date first and define how multi-item delivery dates are handled.

6. Dashboard Summary

The Power BI dashboard includes revenue, order count, AOV, delivered/cancellation/return rates, a yearly revenue trend, order-status distribution, revenue by category/gender/state/payment status/payment method, and a Top 10 Products table. Year, Category, and State slicers support interactive exploration. The AP regional finding should be reflected in the narrative only after its exact state-level outputs are confirmed.

7. Recommended Actions

- Investigate cancellation and return reasons and segment them by channel, payment method, and shipping type.
- Review replenishment for products with low stock and meaningful recent sales.
- Evaluate discount campaigns with margin and conversion data, not AOV alone.
- Compare AP category mix with Telangana and other states using the actual output tables.
- Review payment failure/pending outcomes after confirming payment status definitions.
- Track repeat purchase among purchasing customers and separately track registered customers who never ordered.

8. Suggested GitHub Structure

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ecommerce-sales-customer-analytics/
■■■ README.md
■■■ ecommerce_analysis.sql
■■■ Ecommerce_Sales_Customer_Analytics_Corrected_Insights.pdf
■■■ ecommerce_dashboard.pbix
■■■ dashboard_preview.png
```

Publication note: the SQL file contains queries, not their result grids. Values described as previously reported should be checked against the final database outputs before making the repository public. Never publish credentials or private customer information.